Requirements
Hospital Management - Patients Section
- MysteryInc Team -

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Introduction
The system is designed to centralize patient data, medical histories, and prescriptions while integrating external data from ER and Staff modules.

System Actors
The system supports three primary user roles:
Administrator – manages patient records, archives patients, and views system statistics.
Medical Staff – responsible for recording consultations, diagnoses, and medical records.
Pharmacist – responsible for viewing and fulfilling prescriptions issued by medical staff.

Each role has access only to the functionalities relevant to their responsibilities.

0. Role Selection (Main Window)

When the application starts, the system must display a role selection window that allows the user to choose their role within the hospital. **The main window does not require the registration or login of the user.**

The system must provide three role options:
- Administrator
- Medical Staff
- Pharmacist

After selecting a role, the system must redirect the user to the corresponding role interface.

Each of the 3 windows will also contain a “Back to Home” option, which allows the user to return to the main window and choose another user role.

1. Administrator Dashboard

After authentication, the system must display an administrator dashboard that provides access to the following functionalities:
- View all patients
- View archived patients
- View general statistics

The administrator must be able to navigate to any of these sections from the dashboard.

1.1 View All Patients

The system must allow the administrator to view a list containing all registered patients.

The patient list must display the following basic information for each patient:

- First name
- Last name
- Date of birth
- Sex
- Phone number
- Emergency contact

This information must be presented in a structured table format.

1.1.1 Filter Patients

The system must allow the administrator to filter the patient list using the following criteria:
- Age (computed based on DOB)
- Sex

After applying a filter, the system must display only the patients that match the selected criteria.

The filtered results must be displayed in the patient list.

1.1.2 Search Patients

The system must allow the administrator to dynamically search the patient list using the following criteria:
- First name
- Last name
- CNP

After applying the search, the system must display only the patients that match the selected criteria search query.

The results of the search must be displayed in the patient list.

The system must allow the administrator to combine search and filter criteria. The patient list should dynamically update to display only those individuals who match both the specific search terms and the selected demographic filters

1.1.3 Add Patient

The system must allow the administrator to register a new patient.

When the administrator selects the Add Patient option, the system must display a form containing the following fields:
- First name, represented as a string of a maximum of 100 chars, alphabetical chars, hyphen and space permitted
- Last name, represented as a string of a maximum of 100 chars, alphabetical chars, hyphen and space permitted

- Date of birth, which follows the DD-MM-YYYY format
- Sex, a case-insensitive char of either “F” or “M”
- Phone number, a 9 digits number in the romanian format, prefixed with ‘+40’
- Emergency contact, represented by a 9 digits phone number, prefixed with ‘+40’
- CNP, a 13 digits number that follows the Romanian Personal Identification format

After submitting the form, the system must store the patient information in the database.

The system must validate the input data before storing it. If any required field is missing or incorrectly formatted, the system must display an error message corresponding to the invalid data fields and prevent the operation from being completed.

A verification based on matching the CNP with sex and DOB should also be made.

If a patient with the same identification data represented by the CNP already exists in the system, the system must display a warning message indicating that the patient already exists.

1.1.3.1 Add Patient Medical Information

When a new patient is added, the system must create its medical history, requiring the user to input the following specific medical data (in a new window):

- Blood Type, a case-insensitive string with values in O, A, B or AB
- RH, a case-insensitive string with values of either ‘Positive’ or ‘Negative’
- Chronic Conditions, a string of a maximum 2000 chars
- Allergies: Allergy Name, a string of maximum 100 chars
  Type, a string of maximum 100 chars
  Level, dropdown list containing the strings “Mild”, “Moderate”, “Severe” or “Anaphylactic”
  Category, a string of maximum 100 chars

Only after Step 1 is validated, the system advances to the Medical History view.

1.1.4 Archive Patient (one click)

The system must allow the administrator to archive a patient.

Archiving a patient may occur in special situations such as when the patient is deceased or when the patient should no longer appear in the active patient list.

When the administrator selects the Archive Patient option, the system must display a confirmation dialog requesting confirmation of the action.

If the administrator confirms the action, the patient must be moved from the active patient list to the archived patients list.

Archived patients must no longer appear in the list of active patients.

1.1.5 Update Patient Information (one click)

The system must allow the administrator to update patient information.

When the administrator selects a patient and chooses the update option, the system must display a form containing the existing patient information.

The administrator must be able to modify the following fields:

- First name, represented as a string of a maximum of 100 chars, alphabetical chars, hyphen and space permitted
- Last name, represented as a string of a maximum of 100 chars, alphabetical chars, hyphen and space permitted
- Date of birth, which follows the DD-MM-YYYY format
- Sex, a case-insensitive char of either “F” or “M”
- Phone number, a 9 digits number in the romanian format, prefixed with ‘+40’
- Emergency contact, represented by a 9 digits phone number, prefixed with ‘+40’
- CNP, a 13 digits number that follows the Romanian Personal Identification format

After submitting the changes, the system must update the stored patient information.
Input validation must be applied to ensure that all required fields contain valid data.
A verification based on matching the CNP with sex and DOB should also be made.

1.1.6 View Patient Medical Information (double click)

The system must allow the administrator to view detailed medical information for a selected patient.

When a patient is selected from the patient list, the system must display a detailed view containing the patient’s medical information.

This view must contain two main sections:

- Medical history
- Medical records

1.1.6.1 General Information (Left Section)

The medical history section must display general medical information including:

- Blood type
- RH factor
- Allergies
- Chronic conditions

The system must also display a list with all the medical records associated with the patient.
Each medical record must be identified by its consultation date.

1.1.6.2 Consultation Details (Right Section)

When a medical record is selected, the system must display the following information:

- Patient name
- Consultation source (ER visit or appointment)
- Name of the staff member responsible for the consultation
- Reported symptoms
- Diagnosis
- Consultation date

1.1.6.3 Prescription and Export

The system must also allow the administrator to view the prescription associated with the selected medical record by pressing a button. This will contain:

- Meds list
- Quantity of each medication
- Doctor notes

The administrator must also have the option to export the medical record information as a PDF document in the user's computer.

1.1.6.4 Dynamic Medical Billing & Discount Roulette

Functional Requirement:

Inside each Medical Record, a field labeled "Base Price" will be automatically computed based on the complexity of the consultation. Next to this field, a button "Apply discount" will be available. Clicking this button opens a pop-up window containing a functional Discount Roulette. The final price is calculated only after the spin, and the roulette button is then replaced by the Final Price value.

1. Base Price Scoring Metric (The Complexity Score)

The system computes the initial cost P_base by analyzing the patient's current record and medical history:

- Visit Type (Base Fee):
  - ER Visit: 500 units.
  - Standard Appointment: 200 units.
- Transplant Factor: If the record is linked to a TransplantID, add 2000 units.
- Medication & Prescriptions: +50 units for every item in the PrescriptionItems table.
- Clinical Complexity:
  - Chronic Conditions: +100 units for each condition listed in MedicalHistory.
  - Allergy Severity: Points added based on the SeverityLevel in PatientAllergies:
    - Mild/Moderate: +20 units.
    - Severe/Anaphylactic: +100 units.

Base Formula: P_base = Cost(Visit) + Cost(Transplant) + (Meds * 50) + (Conditions * 100) + sum(AllergyPoints)

2. The Discount Roulette
Implementation Details:

- The Pop-up: When the doctor or admin clicks the roulette icon, a graphical wheel appears with segments for: 0% (No Discount), 10%, 25%, 50%, and 100% (Free of Charge).
- The Spin: The user clicks "SPIN". An animation plays, and the system selects a random discount D.
- Final Calculation: The system applies the discount to the base price: P_final = P_base * (1 - D)
- UI Update: Once the result is determined, the Roulette Pop-up closes. In the Medical Record view, the Roulette Button disappears and is replaced by the P_final value.

1.2 Archived Patients

The system must allow the administrator to view a list of archived patients.

This list must contain the same basic patient information as the active patient list:

- First name
- Last name
- Date of birth
- Sex
- Phone number
- Emergency Contact
- CNP

The system must also allow the administrator to restore an archived patient.

When the Dearchive Patient option is selected, the patient must be moved back to the active patient list and removed from the archived patients list.

1.2.1 Archive Patient as Deceased

The system must allow the admin to mark a patient as dead. The date of death will be saved in the database and the admin would not be able to dearchive it. When a patient is marked as deceased, a pop-up appears stating “This patient has now become a ghost. Beware!!!”

1.2.2 Post-Mortem Organ Donation

When a patient is marked as deceased, a button labeled “Mark as Organ Donor” will appear. By pressing this button, a pop-up window opens where the Admin selects the organ available for donation (e.g. heart, kidney, liver). Once the organ is selected, a table will be displayed listing the top 5 most compatible recipients from the active waitlist, ranked by their compatibility score.

Scoring Metric:

- Hard Filters: If a recipient fails any of these, they are excluded from the match:
    - Blood Type Compatibility: Must follow the organ-specific compatibility matrix.
    - Chronic Conditions: If the recipient has a history of chronic conditions that prevent transplantation, they are excluded.
- Points (Max 100):
    - Blood Type Match (50/25 points): Identical match (e.g. A+ to A+) gets 50 points. Compatible match (e.g., O- to A+) gets 25 points.
    - Age Proximity (30 points): Calculated using DateOfBirth. A 5-year gap or less gets 30 points. The score decreases by 5 points for every additional 5-year gap (10-year gap: 25 points, 15-year gap: 20 points, etc.).
    - Medical Urgency (20/5 points): Based on the High-Risk logic. If the patient has 10+ ER visits in the last 3 months, they receive 20 points. Otherwise, they receive 5 points.

Formula: FinalScore = Score(Blood) + Score(Age) + Score(Urgency)

1.2.3 Ghost Sighting Interaction

A new button labeled "I saw a ghost!" must be placed immediately adjacent to the "Back to Home" button on all main windows (Admin, Staff, and Pharmacist). The system must monitor the frequency of ghost sightings. If the number of "I saw a ghost!" button clicks within a single 24-hour period is larger than 3, the system must trigger a critical intervention. The ghost sighting counter must be globally shared across all three roles (Admin, Staff, Pharmacist) so that 3 total clicks from any user triggers the alert.

The “Exorcism” Alert: A full-screen, non-dismissible alert will pop up with the following message: “CRITICAL PARANORMAL ACTIVITY DETECTED: Multiple sightings confirmed. Please CALL THE PRIEST immediately.”

1.3 General Statistics

The system must provide a statistics section that displays aggregated information about patients and consultations. On the left there will be a menu with all the options, and on the right there will be displayed the desired statistics that will be clicked.

The statistics section must include the following information:

- Total number of patients, including a graphical representation of active and archived patients, in the form of a pie chart
- Distribution of consultations based on source (ER visits or appointments) in the form of a pie chart
- A top 10 of most common diagnoses
- A list of the top 20 most prescribed medications, ordered decreasingly by the number of times they have been prescribed
- Patient age distribution, in the form of a bar chart
- Patient sex distribution, in the form of a bar chart

- Number of consultations over a period of a month, 3 months and a year, in the form of a bar chart

2. Medical Staff Dashboard

After authentication, the system must display a Medical Staff dashboard that allows personnel to manage patient consultations and access medical histories.

The interface must feature a primary search utility at the top of the page, consisting of a search bar and a “Go” action button.

2.1 Patient Search

The system must allow medical staff to search for a specific patient using the following criteria:

- Patient Name
- Personal Identification Number (CNP)

2.1.1 Search Outcomes

- No Match: If no patient records correspond to the input, the system must display an exception message: “There are no patients with this name or CNP.”
- Match Found: If matches exist, the system must display a results table containing:
    - First Name
    - Last Name
    - CNP
    - Date of Birth (DOB)

2.2 Patient Medical Profile

The system must allow the user to select a patient from the search results to open a detailed medical profile window. This window must be organized into a split-screen layout.

2.2.1 General Information (Left Section)

The left section must display a structured table of the patient’s permanent medical data:

- Blood Type and RH Factor
- Allergies
- Chronic Conditions
- File Records: A collapsible list of all previous consultations, labeled by their respective dates.

2.2.2 Consultation Details (Right Section)

When a specific row is selected from the File Records list (double-click), the right section must populate with the full medical record, including:

- Patient Name
- Source (ER or Appointment)
- Staff ID
- Reported Symptoms
- Diagnosis
- Consultation Date

2.2.3 Prescription and Export

For the selected medical record, the system must provide the following functional options:

- View Prescription: Display the associated medication list and dosage instructions.
- Export to PDF: Generate a downloadable PDF file of the medical record for the user's local storage.

2.2.4 Add New Record (Data Import)

The system must provide an "Add a New Record" button at the top-right of the patient profile. Upon selection, the system will initialize a new consultation form by importing data based on the patient's arrival source:

Case A: Import from Staff (Standard Appointment) (Staff = Dev Core Team)

If the record source is an Appointment, the system shall automatically import the complete dataset from the Hospital Staff Module:

- Source ID & Staff ID
- Reported Symptoms
- Diagnosis
- Consultation Date
- Prescription Details: Medication List, Quantities, and Doctor's Notes.
- Entry Date (Timestamp of the record creation).

Case B: Import from ER & Staff (Emergency Visit) (ER = House MD Team)

If the record source is an ER Visit, the system must perform a hybrid import from both modules to consolidate the record:

- From ER Team Module: The system imports the initial contact data:
    - Source ID & Staff ID
    - Reported Symptoms
    - Consultation Date

- From Staff Team Module: The system imports the clinical resolution data:
    - Diagnosis
    - Prescription Details: Medication List, Quantities, and Doctor's Notes.
    - Record Date

After the import is triggered, the medical staff member reviews the consolidated data. Upon clicking "Submit," the record is saved to the central database and appended to the patient's File Records history.

2.2.5 Compatible Blood Donors

When one of the patients is clicked on (one click) a button labeled "Compatible Blood Donors" will appear. By pressing this said button, a new window will be opened. It will contain a table in which there will be listed the first 20 most compatible patients based on the score. The score is calculated based on these criteria:

- blood type and Rh
- chronic conditions or severe allergies
- age range
- sex

Scoring metric:

Hard Filters: If a donor fails any of these, their score is irrelevant because they cannot donate to this specific patient

- Blood Type & Rh factor: Must follow the compatibility matrix (e.g., O- can give to anyone, but AB+ can only give to AB+).
- Chronic conditions: If "true", exclude immediately
- Severe allergies: If "true", exclude immediately

Points: For the donors who passed the hard filters, we calculate a score out of 100 points to rank the top 20.

- Blood Type Match (50/25 points): Identical match(e.g. A+ to A+) gets full points. Compatible match (e.g. O- to A+) gets partial points
- Age Proximity (30 points): Donors in the same age group (max 5 year gap) get full points. Points decrease by 5 for every increase in the age group (e.g. 10 year gap: 25 points, 15 year gap: 20 points, etc.).
- Sex Match (20/10 points): Same sex gets full points (often preferred in specific transplants/transfusions to reduce minor antigen reactions). Different sex gets partial score.

The final score for each patient will be computed using the formula: FinalScore = Score(Blood) + Score(Age) + Score(Sex).

The table will have the following columns for each compatible patient:

- first name

- last name
- CNP
- blood type
- rH
- score

2.2.6 High Risk Patients Alert

After a patient is selected from the table, the doctor should be able to see on the patient's specific page, an alert pop up that informs him if the patient is a high risk one, meaning he has more than 10 ER visits in the last 3 months.

2.2.7 Transplant Waitlist Request (Medical Staff Module)

In the Patient Profile view, next to the "Compatible Blood Donors" button, a new button labeled "Request Organ Transplant" will be available. Pressing this button opens a pop-up window where the doctor can input the specific OrganType (e.g. heart, kidney, liver) required by the patient. Upon submission, a new entry is created in the Transplants table with the status 'Waiting'.

Implementation Details:

- Waitlist Entry: The system creates a record storing the ReceiverID (current patient), the OrganType, and the RequestDate. The DonorID remains NULL, indicating the patient is active on the Virtual Waitlist.
- Urgency & Eligibility Check: Before finalizing the request, the system performs a real-time check:
    - Urgency: If the patient has 10+ ER visits (High-Risk), a red "Urgent Priority" label is attached to the request.
    - Warnings: If ChronicConditions exist in MedicalHistory, the system displays a warning: *"Patient has underlying conditions that may affect transplant success."*
- Live Ranking: Once on the waitlist, the patient's data (Blood Type, Age, ER visits) is ready to be compared against any future deceased donor.
- Status Visibility: The doctor can monitor the Status field. It remains 'Waiting' until the Administrator finds a match and assigns a DonorID, at which point the status changes to 'Scheduled'.

3. Pharmacist Dashboard

The Pharmacist serves as the final link in the medical chain, responsible for verifying and fulfilling medication orders. Upon entry, the Pharmacist is presented with a comprehensive feed of all prescriptions issued within the system.

3.1 Prescription View

The main view consists of a chronological feed of prescriptions, arranged in descending order (most recent first). To ensure clarity, each prescription is represented as a card, placed side by side, 3 prescriptions on each row. By default, the feed displays only the last 9 prescriptions issued to ensure optimal performance, thus we will have a 3 by 3 grid on each page. The pharmacist can use navigation buttons (Next/Previous) to view older records.

Contents of the prescription card:

- Prescription ID: Unique identifier for the medication order. (placed on the middle top of the card) - clickable item that will open the meds list section.
- Patient Name: The person for whom the prescription is.
- Doctor Name: The staff member who authorized the prescription.
- Doctor Notes: Specific instructions regarding dosage or administration.
- Date: The exact timestamp of when the prescription was issued.

When the specific Prescription ID is clicked, a pop up message containing the Meds list(a list with medicine names and prescribed quantity for each name, placed on a row each) will be shown. The pop up message can then be closed, allowing the user to fully see the main prescription view again.

3.2 Filter Prescriptions

To manage the high volume of data, a specialized filter bar is provided at the top of the page. The Pharmacist can narrow down the list using the following criteria:

- ID Search: Filter by specific Prescription ID.
- Name Search: Filter by specific Patient/Doctor Name.
- Medication Search: Filter by specific drug names within the Meds List.
- Temporal Search: Filter by specific dates or date ranges.

3.3 Anti-Drug-Shopping & Addict Detection

The Pharmacist Dashboard will feature a specialized monitoring tool labeled “See All Addicts.” Clicking this button triggers a database-wide scan to identify patients exhibiting "Drug Shopping" behavior. Flagged patients are displayed in a high-priority list, allowing the pharmacist to intervene before dispensing medication.

1. Addict Detection Logic

The system automatically identifies a patient as an "Addict Candidate" if the following cumulative criteria are met within the last 30 days:

- Frequency: The patient has been issued more than 3 separate prescriptions.
- Consistency: All prescriptions are for the same medication name (identified via PrescriptionItems).

- Source Diversity: The prescriptions were issued by different doctors (distinct StaffID entries in the MedicalRecord table).

2. Pharmacist Oversight & Law Enforcement Integration

For every flagged patient, the pharmacist is provided with the following decision-making tools:

- Chronic Conditions Access: A "View History" button that displays the patient's MedicalHistory.ChronicConditions. This allows the pharmacist to verify if the high volume of medication is clinically justified (e.g., chronic pain management).
- Law Enforcement Alert: A high-visibility "Notify Police" button. Upon clicking, a confirmation pop-up is generated, documenting the 3 suspicious medical records and sending a simulated alert to local authorities.

Non-Functional Requirements

The system must store data persistently using a relational database.
The system must validate all user input before storing information in the database.
The system must maintain consistency between patient records, medical histories, and prescriptions.
The system must support exporting medical records in PDF format.